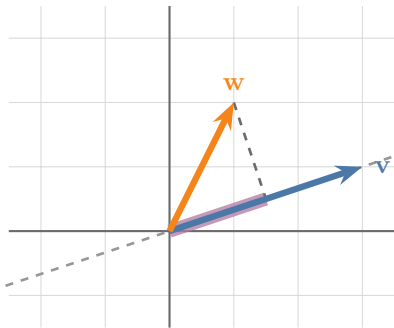
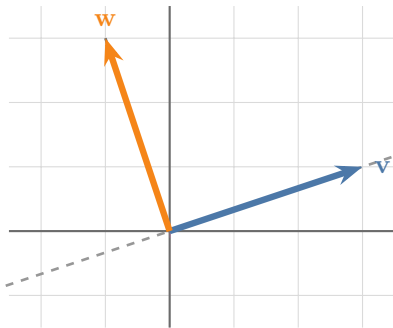


Same general direction



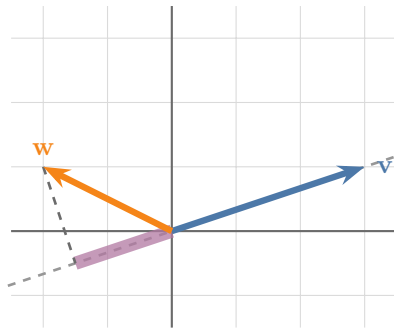
$$\mathbf{v} \cdot \mathbf{w} = 1.58 \times 3.16 = +5$$

Perpendicular



$$\mathbf{v} \cdot \mathbf{w} = 0 \times 3.16 = 0$$

Opposite general direction



$$\mathbf{v} \cdot \mathbf{w} = -1.58 \times 3.16 = -5$$

purple: the projection of $\mathbf{w}$ onto the line of $\mathbf{v} = [3, 1]$; dot product = (signed length of the projection) $\times$ (length of $\mathbf{v}$)